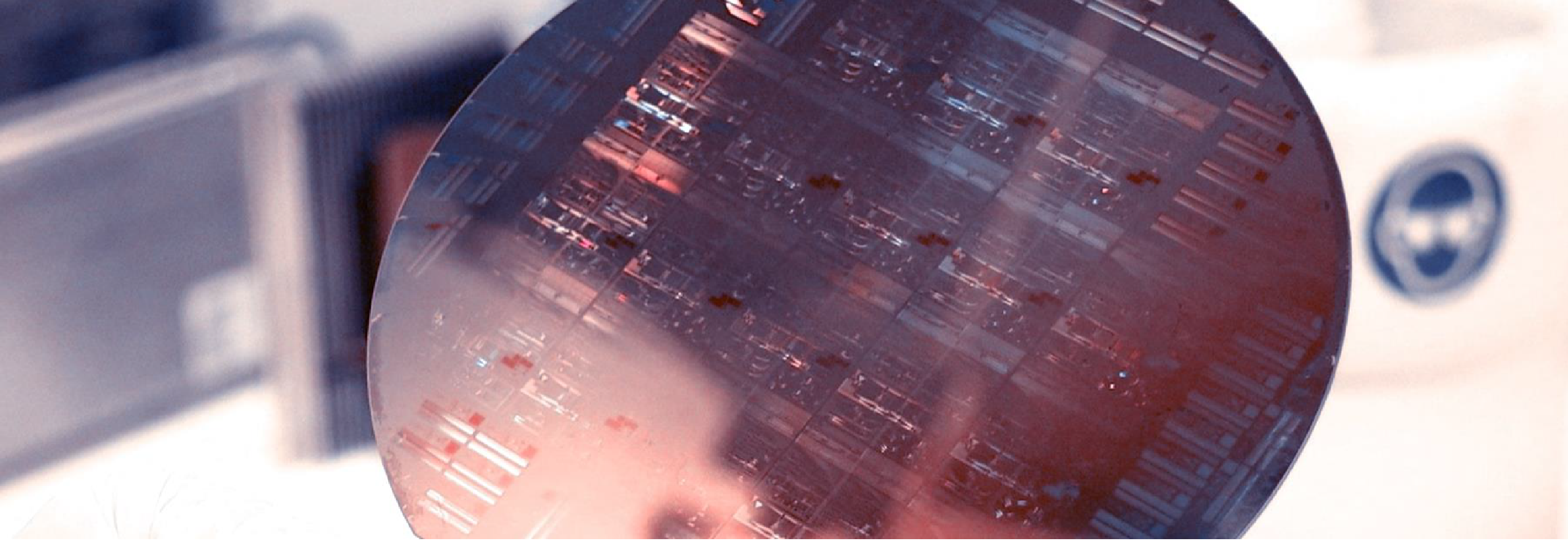


PIC-UPV
PERTE CHIP CHAIR

Lab Book 0.3. Ring Resonator concepts

Student:

Instructions

General

SCRIPTS AND WORKFLOW

1. Fork / clone the repository: https://github.com/cccanvas-upv/pic-upv-lab0.git

2. Open the repository in VS Code with WSL; run uv sync if needed.

3. Select the project .venv as your Jupyter kernel.

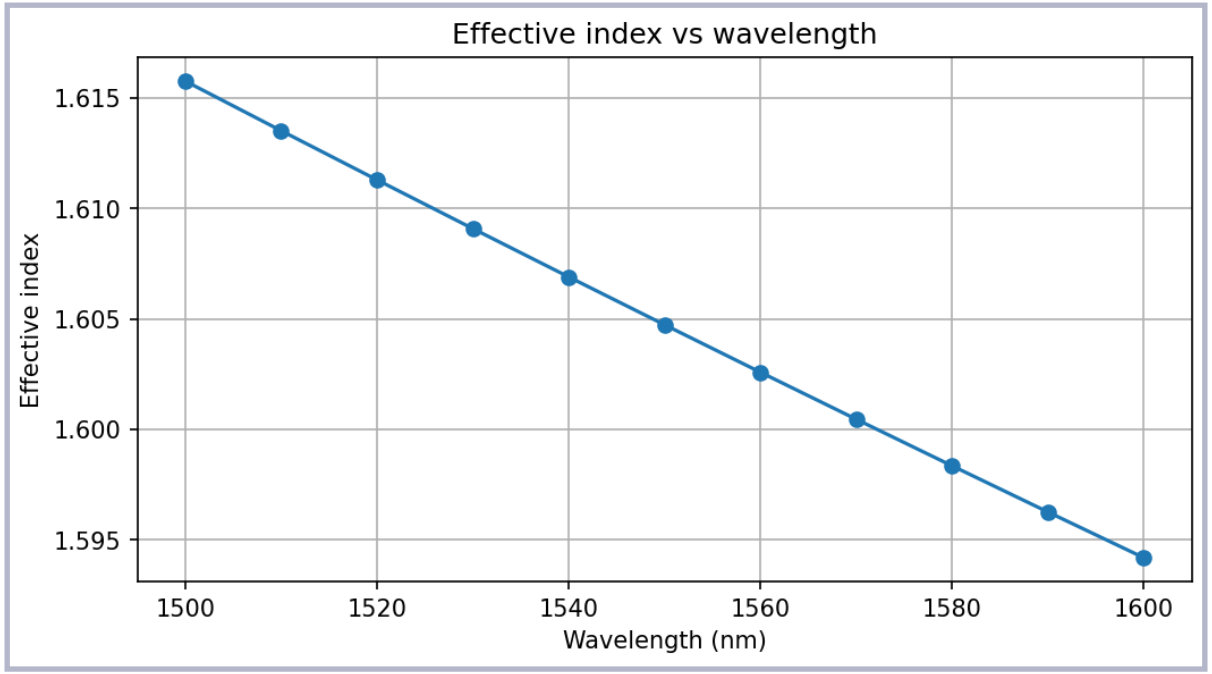
4. Run 030_RING_RESONATOR_Student.ipynb in sequence.

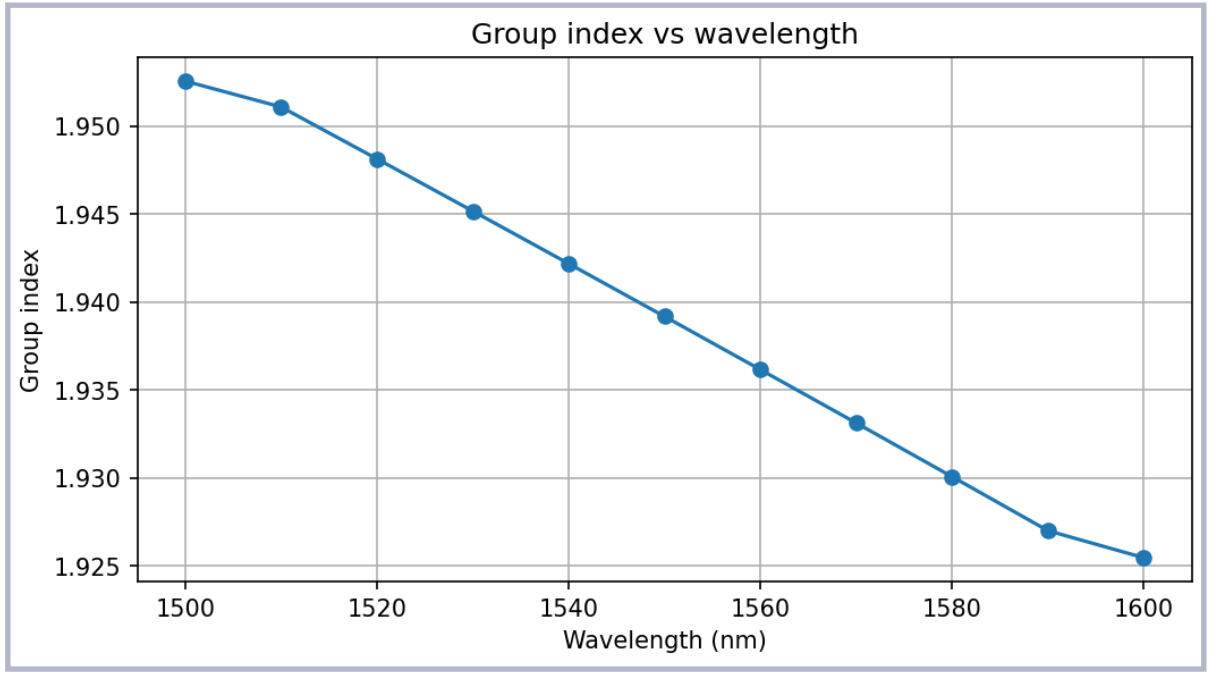
5. Paste the requested plots and fill in your results in this LabBook.

LO.1 – Effective and Group Index Analysis

Mode solver sweep: 1500–1600 nm (11 wavelengths), fundamental TE-like mode

At 1550 nm: **n_eff** = 1.60473 **n_g** = 1.93918





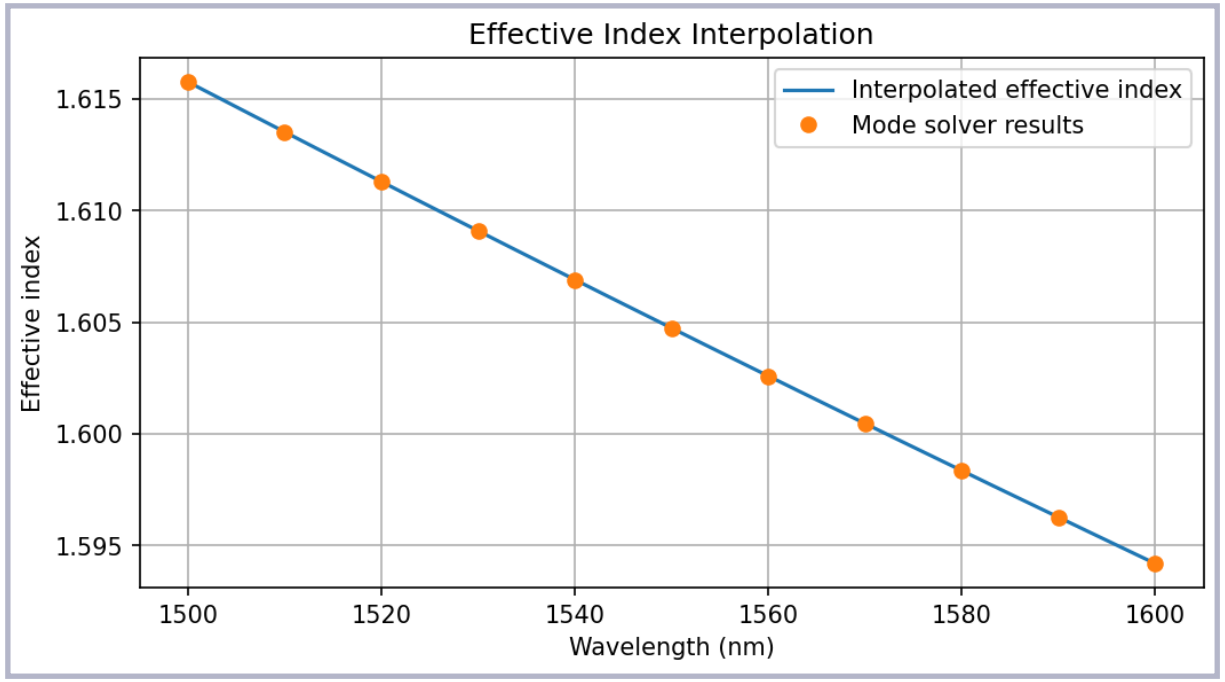
Refractive indices describe the ratio of the speed of light in a vacuum to that in a medium. The effective refractive index, n_{eff}, is the refractive index of a waveguide for a given wavelength and mode; as well as the wavelength and mode of the signal, the cladding and waveguide material, geometry and dimensions are all key factors to determining the effective index. The group index is the ratio of the speed of light in a vacuum to the group velocity of the light passing through the waveguide.

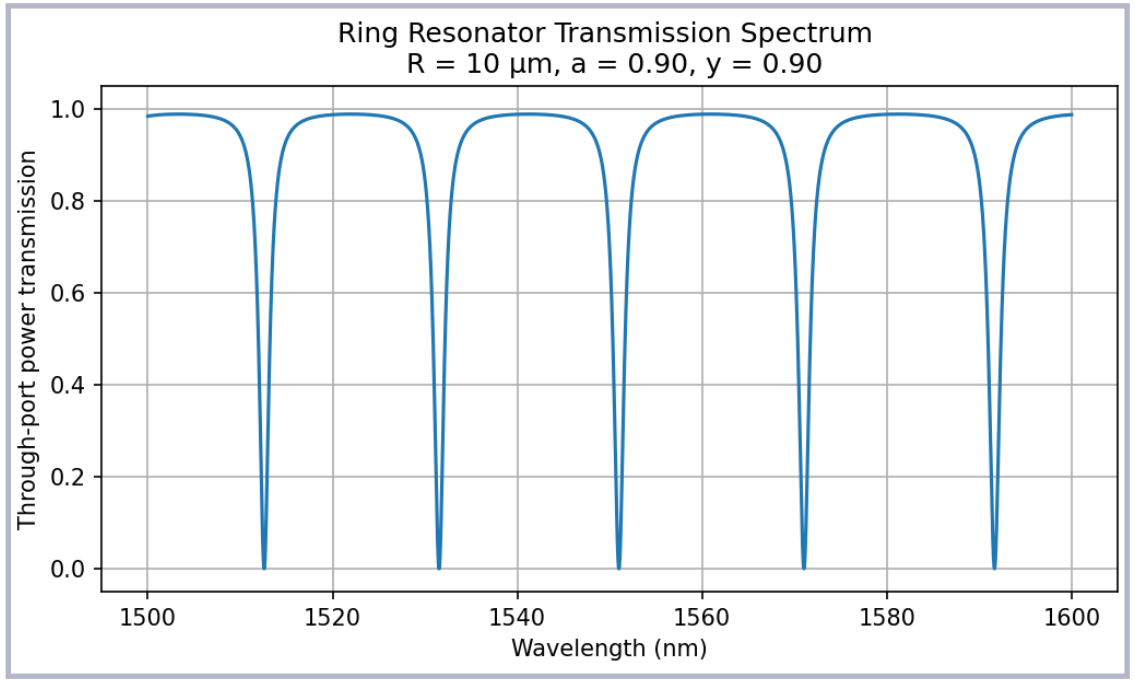
The plots show while both n_{eff} and n_g decrease with respect to wavelength, it s while n_g decreases with wavelength with a non-linear relationship. This is because the effective index is linearly dependent on the wavelength, $n_{eff} = \frac{\beta\lambda}{2\pi} \quad \rightarrow \quad n_{eff} \propto \lambda.$

Whereas, the n_g is related to the wavelength and the differential of n_{eff}, $n_g = n_{eff} - \lambda\frac{dn_{eff}}{d\lambda} \quad \rightarrow \quad n_g \propto \lambda\frac{dn_{eff}}{d\lambda}.$

LO.2 – Ring Resonator Spectral Response

Interpolate n_eff(λ), then calculate the through-port spectrum of an all-pass ring (R = 10 μm; a = y = 0.90).

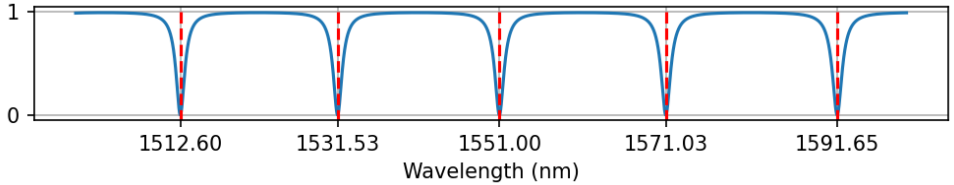




List the resonance wavelengths and estimate the FSR near 1550 nm.

From the ring resonator transmission spectrum, the resonant wavelengths can be extracted: 1512.60, 1531.53, 1551.00, 1571.03, 1591.65nm

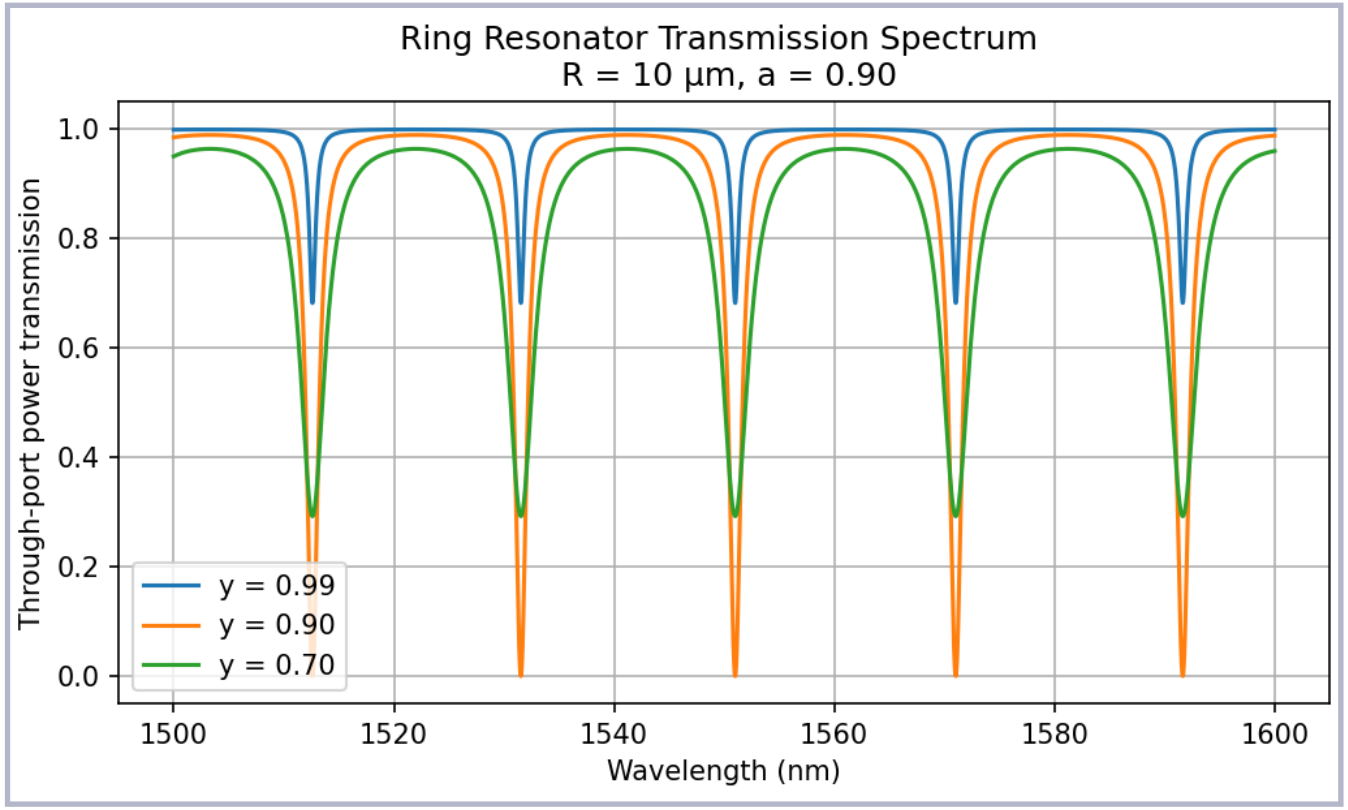
And an estimated FSR around wavelength 1550nm: 20nm



Explain why transmission reaches a minimum at resonance and approaches unity off resonance.

The transmission falls to a minimum at wavelengths that successfully couple to the ring and fall in phase causing the resonance. All other frequencies don't couple with the ring resonator, meaning the optical power at those frequencies transmit almost almost entirely with no loss.

LO.3.1 – Influence of Coupling



The transmission depth at resonance peaks only fall to zero when the coupling ratios are equal, a=y=0.9 (critical coupling). The figure shows that if the system is under-coupled (a < y=0.99), the resonant peak is very sharp and narrow (small FWHM) but the peak itself isn't anywhere near zero, meaning most of the light isn't coupling to the ring and destructively interfering.

However, when the system is over-coupled (a > y = 0.7) the peak is deeper than that of the under-coupled system but still not falling to zero, and the shape of the peak is very broad, indicating that the ring is coupling to more wavelengths causing some power loss across the spectrum, but isn't completely filtering out the resonant frequencies.

y	Regime
0.99	Undercoupled
0.90	Critical
0.70	Overcoupled

Which condition produces the deepest dip?

Critical coupling produces the deepest dip, which is when *a* and *y* are equal. In this case, when a=y=0.9.

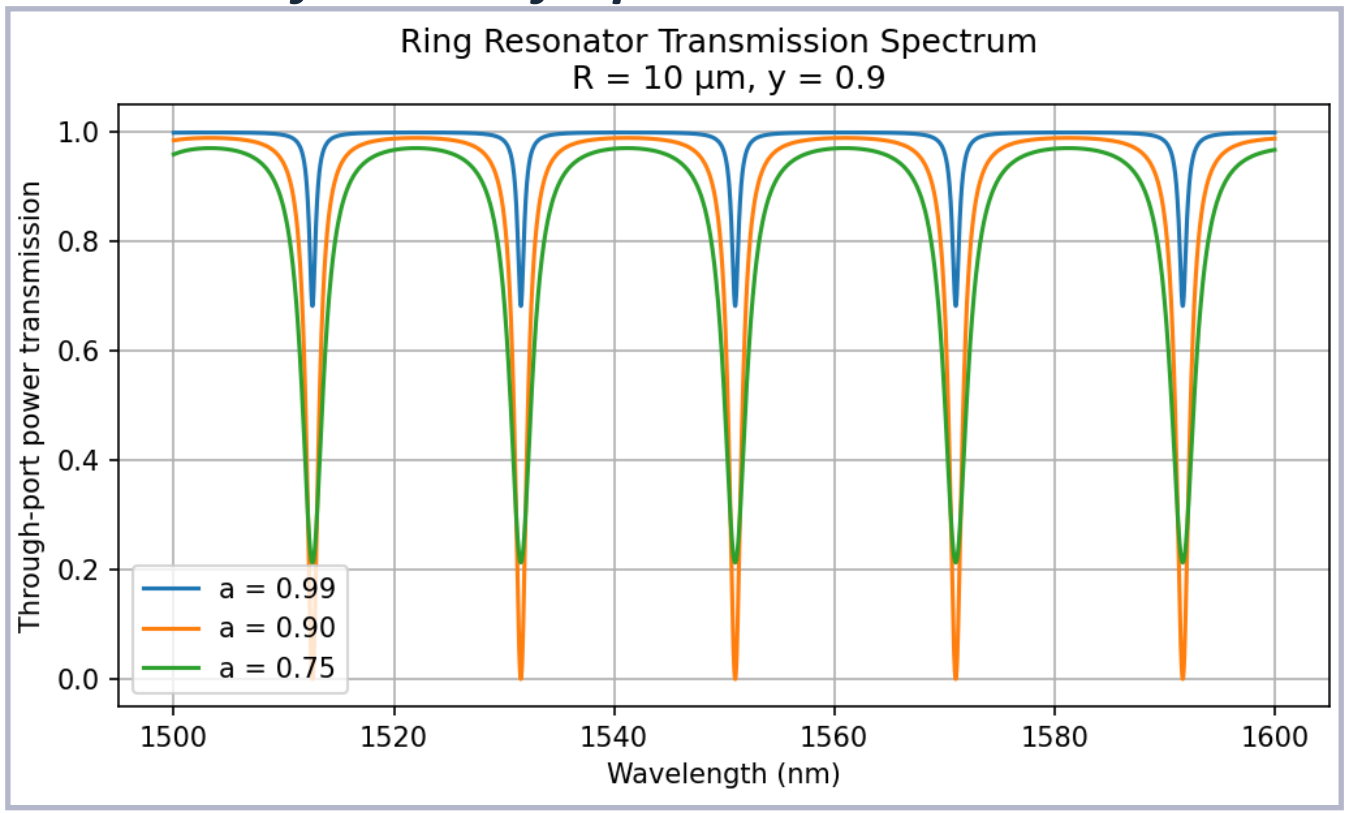
Why does critical coupling ideally suppress through-port transmission?

The power losses from the resonator match that of the coupled signal power, meaning the signal destructively interferes perfectly.

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TSI-069100-2023-0002

LO.3.1 – Influence of Optical Losses

Exercise 2 · Fix R = 10 μm and y = 0.90. Compare a = 0.99, 0.90 and 0.75.



a	Coupling regime
0.99	Overcoupled
0.90	Critical
0.75	Undercoupled

Questions:

1. *How does increasing the propagation loss affect the resonance depth?*

   If the propagation loss coefficient is too high, the resonant depth doesn't fall to zero, similarly with when the loss coefficient is too low.

2. *Which configuration satisfies the critical coupling condition?*

   The critical coupling condition is met when the power loss coefficient and the coupling coefficient match (i.e. *a==y)*

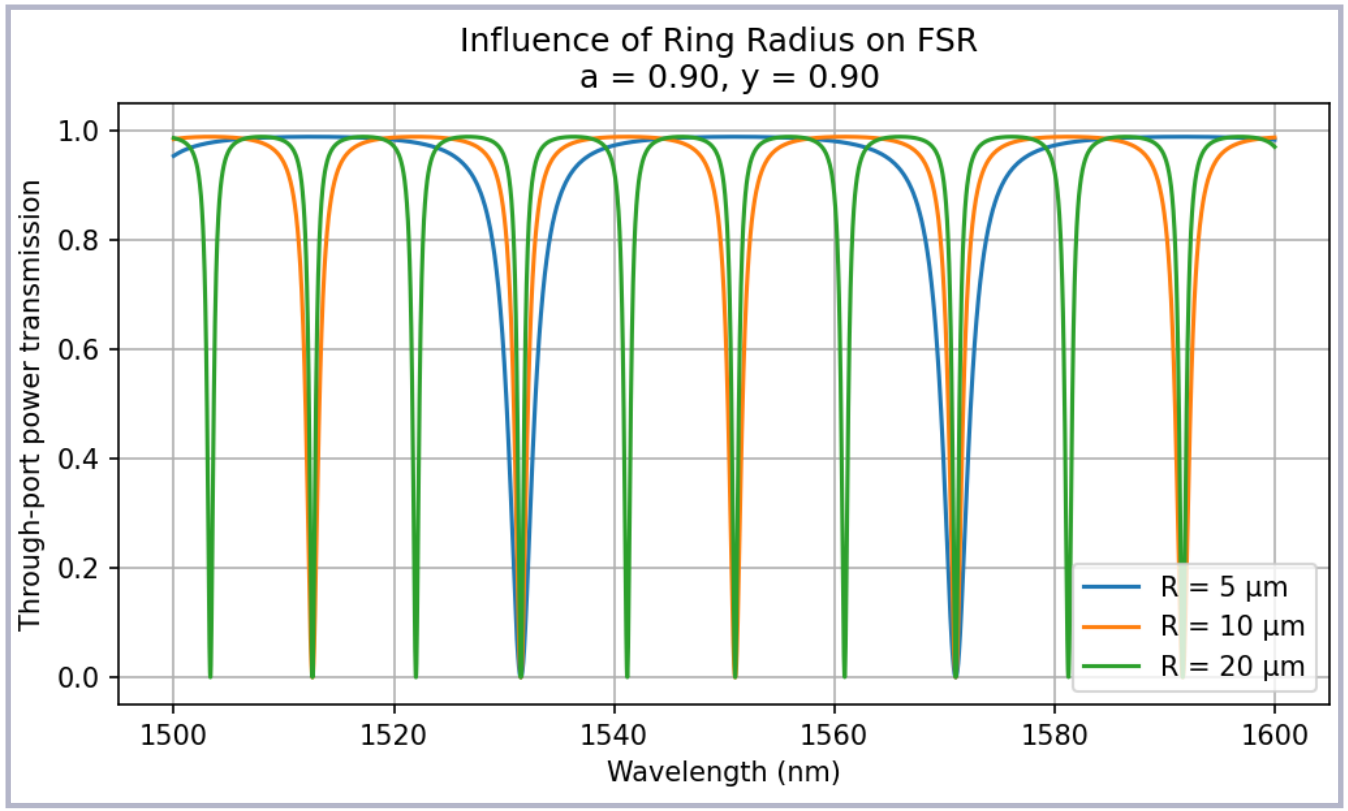
3. *Can increasing the internal losses produce a deeper resonance? Explain why.*

   Increasing the resonator losses beyond the critical coupling condition won't produce a deeper resonance. If the internal losses are greater than the coupled power, then the signal amplitude within the ring won't be great enough to completely destructively interfere, which means that the resonance won't get any deeper.

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LO.3.2 – Ring Radius and FSR

Exercise 3 · Fix a = y = 0.90; compare radii R = 5, 10 and 20 µm.



R (µm)	Sim. FSR	Theory FSR
5	39.50	39.44
10	19.76	19.72
20	9.813	9.86

Questions:

1. *How does the FSR change when the ring radius increases?*

   The FSR decreases with increasing ring radius

2. *Explain physically why the FSR depends on the ring circumference.*

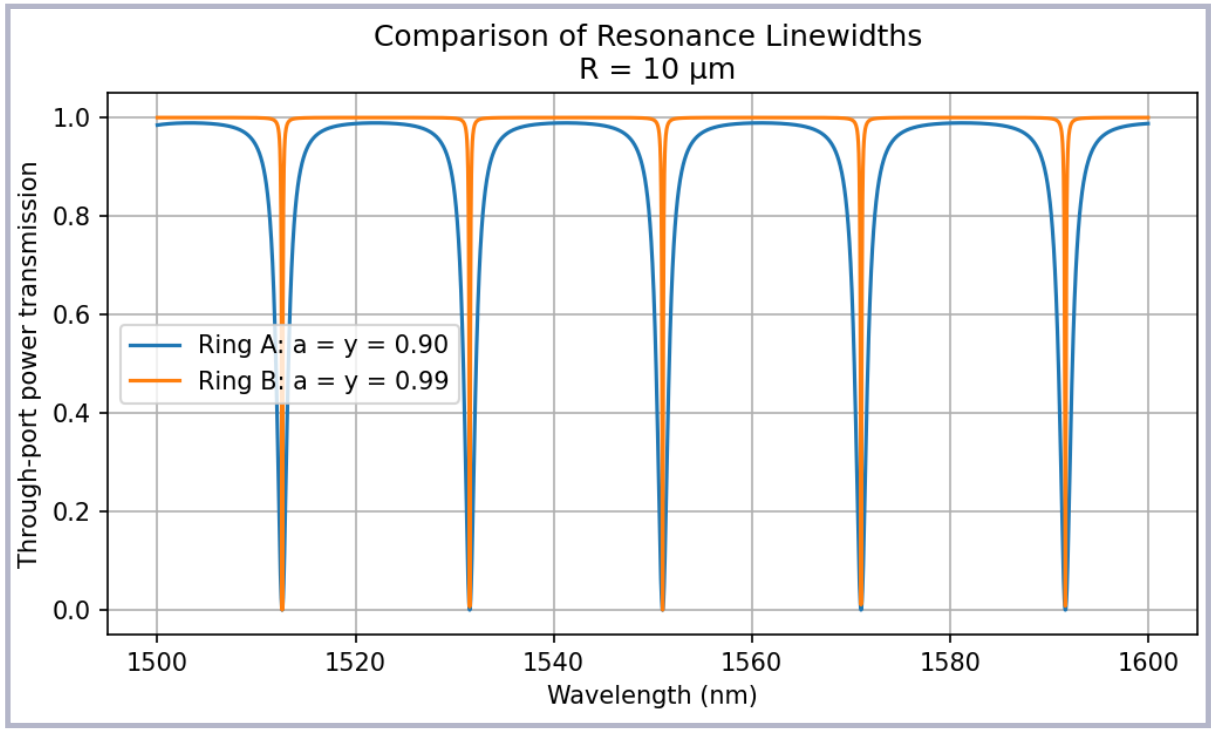
   The FSR is the frequency range between two resonance points, and resonance is dependent on the phase difference of the wave traveling round the length of the ring. So if the path length is longer, it means there are more wavelengths that reach a 2pi phase difference, creating the resonant peaks.

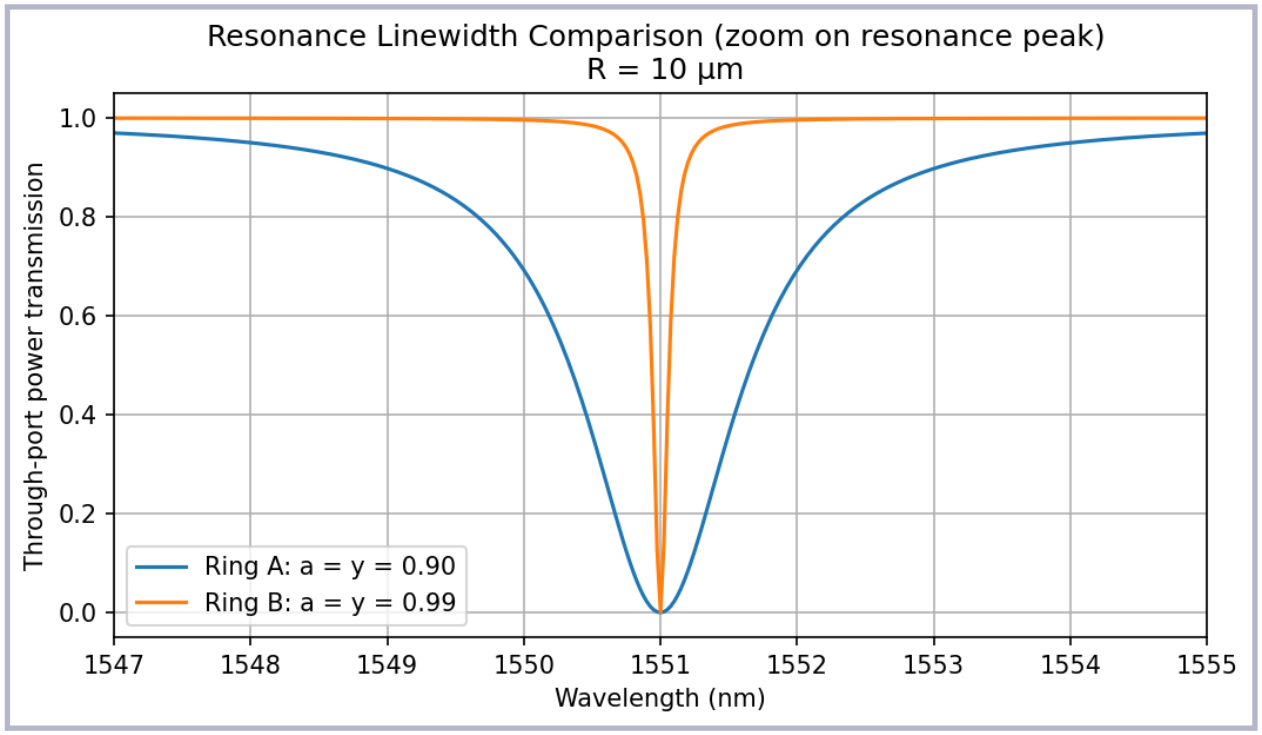
3. *Why is the FSR not exactly constant across the wavelength range?*

   Due to the inverse relationship between frequency and wavelength. And the FSR is proportional to the square of the wavelength.

LO.3.3 – Resonance Linewidth

Exercise 4 · Ring A: a = y = 0.90; Ring B: a = y = 0.99. Both have R = 10 µm.





Estimate the FWHM graphically at half the dip depth (T ≈ 0.5). No extra simulations required.

Ring	Estimated FWHM (nm)
A	1.3nm
B	0.2nm

LO.3.3 – Finesse, Q-factor and Applications

Use the FSR from Exercise 3 and the visually estimated FWHM from Exercise 4.

Parameter	Ring A	Ring B
Resonance wavelength (nm)	1551	1551
FSR (nm)	19.475	19.475
Estimated FWHM (nm)	1.3nm	0.2nm
Finesse	14.98	97.375
Loaded Q-factor	1193.1	7755

Discussion.

1. Which ring has a narrower resonance and higher Q? Why?

Ring B

2. Which is preferable for detecting small resonance shifts in sensing?

3. Is a narrower resonance always better for WDM filtering? Discuss bandwidth and stability.